

Solving SAS and SSS Triangles with the Law of Cosines

Answer Key

Precalculus

Quick Answers

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|------------------|------------------------------------|
| 1. 7 | 7. 59.49 degrees |
| 2. 81.79 degrees | 8. 48.31 degrees |
| 3. 8.62 | 9. (a) angle A = 102.64 degrees, — |
| 4. 36.87 degrees | 10. 98.51 m |
| 5. 25.07 degrees | 11. 87.27 degrees |
| 6. 24.28 | 12. (a) distance c = 169.60 m, — |
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What is verified

12 of 14 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 9, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSG-SRT.D.10 – *problems 1, 2, 3, 4, 6, 7, 9*

HSG-SRT.D.11 – *problems 5, 8, 10, 11, 12*

Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

- If they got 9.43: forgot the middle term and treated the triangle as right-angled*
 - If they got 81.76: used the reference angle of forty-eight degrees instead of the obtuse angle, which flips the sign of the middle term*
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- 1.** $a = 8$, $b = 5$, $C = 60^\circ$. Find c .

Solution: two sides and the angle between them — SAS — so use the law of cosines with c on the left, because C is the known angle.

$$c^2 = 8^2 + 5^2 - 2(8)(5) \cos 60^\circ = 64 + 25 - 80 \left(\frac{1}{2}\right) = 49$$

$c = 7$

Common wrong answer: 9.43, from dropping the $-2ab \cos C$ term and using $\sqrt{8^2 + 5^2}$. That is the Pythagorean theorem, which is the law of cosines only when $C = 90^\circ$.

2. $a = 5, b = 7, c = 8$. Find C .

Solution: three sides — SSS — so rearrange for the cosine of the angle opposite c .

$$\cos C = \frac{5^2 + 7^2 - 8^2}{2(5)(7)} = \frac{25 + 49 - 64}{70} = \frac{10}{70} = 0.142857\dots$$

$$C = \cos^{-1}(0.142857\dots) = 81.7868\dots^\circ$$

$C \approx 81.79^\circ$

3. $b = 11, c = 14, A = 38^\circ$. Find a .

Solution: A lies between b and c , so put a on the left.

$$a^2 = 11^2 + 14^2 - 2(11)(14) \cos 38^\circ = 121 + 196 - 308(0.7880\dots) = 74.3272\dots$$

$$a = \sqrt{74.3272\dots} = 8.6193\dots$$

$a \approx 8.62$

4. $a = 9, b = 12, c = 15$. Find A .

Solution: SSS again, this time solving for the angle opposite a — so a^2 is the term that gets subtracted.

$$\cos A = \frac{12^2 + 15^2 - 9^2}{2(12)(15)} = \frac{144 + 225 - 81}{360} = \frac{288}{360} = 0.8$$

$A = \cos^{-1}(0.8) = 36.8699\dots^\circ$. Since $9^2 + 12^2 = 15^2$, this triangle is right-angled at C , and the answer is the familiar 36.87° .

$A \approx 36.87^\circ$

5. $a = 6, b = 10, C = 110^\circ$. Find A .

Solution: no side has a known opposite angle yet, so the law of cosines has to run first.

$$c^2 = 6^2 + 10^2 - 2(6)(10) \cos 110^\circ = 136 + 120(0.3420\dots) = 177.0424\dots, \quad c = 13.3057\dots$$

Now c and C are a pair, so the law of sines finishes it:

$$\sin A = \frac{6 \sin 110^\circ}{13.3057\dots} = 0.4236\dots, \quad A = 25.0709\dots^\circ$$

A must be acute here, since C is already obtuse and a triangle has only one obtuse angle.

$A \approx 25.07^\circ$

6. $a = 23, c = 17, B = 73^\circ$. Find b .

Solution: B lies between a and c , so b goes on the left.

$$b^2 = 23^2 + 17^2 - 2(23)(17) \cos 73^\circ = 529 + 289 - 782(0.2923 \dots) = 589.3653 \dots$$

$$b = 24.2768 \dots$$

$$b \approx 24.28$$

7. $a = 13$, $b = 14$, $c = 15$. Find B .

Solution: solve for the angle opposite b , so b^2 is subtracted.

$$\cos B = \frac{13^2 + 15^2 - 14^2}{2(13)(15)} = \frac{169 + 225 - 196}{390} = \frac{198}{390} = 0.5076 \dots$$

$B = 59.4898 \dots^\circ$. As b is the middle-length side, B should be the middle-sized angle, and it is.

$$B \approx 59.49^\circ$$

8. $b = 9$, $c = 12$, $A = 47^\circ$. Find B .

Solution: SAS, so find the third side first.

$$a^2 = 9^2 + 12^2 - 2(9)(12) \cos 47^\circ = 225 - 216(0.6819 \dots) = 77.6884 \dots, \quad a = 8.8140 \dots$$

Then, with a and A a pair,

$$\sin B = \frac{9 \sin 47^\circ}{8.8140 \dots} = 0.7468 \dots, \quad B = 48.3121 \dots^\circ$$

B is acute because b is not the longest side.

$$B \approx 48.31^\circ$$

9. $a = 20$, $b = 16$, $c = 9$. (a) Find A . (b) What did $\cos A$ tell you first?

(a)

$$\cos A = \frac{16^2 + 9^2 - 20^2}{2(16)(9)} = \frac{256 + 81 - 400}{288} = \frac{-63}{288} = -0.21875$$

$$A = \cos^{-1}(-0.21875) = 102.6356 \dots^\circ$$

$$A \approx 102.64^\circ$$

(b) *Instructor-judged.* A correct response reads the sign before computing: the cosine came out negative, and cosine is negative only between 90° and 180° , so A is obtuse. A strong answer adds the structural reason — $a = 20$ is the longest side, so it faces the largest angle, and only the largest angle of a triangle can be obtuse. Reporting the computed value alone is not an explanation.

10. Paths: $a = 120$ m, $b = 95$ m, included angle $C = 53^\circ$. Find c .

Solution: the car park is the vertex between the two paths, so the angle there is included between a and b — SAS.

$$c^2 = 120^2 + 95^2 - 2(120)(95) \cos 53^\circ = 23425 - 22800(0.6018 \dots) = 9703.6 \dots$$

$$c = 98.5069 \dots$$

$c \approx 98.51 \text{ m}$

11. Markers: $a = 340 \text{ m}$, $b = 410 \text{ m}$, $c = 520 \text{ m}$. Find C .

Solution: SSS, solving for the angle opposite the longest side.

$$\cos C = \frac{340^2 + 410^2 - 520^2}{2(340)(410)} = \frac{115600 + 168100 - 270400}{278800} = \frac{13300}{278800} = 0.0477 \dots$$

$C = 87.2657 \dots^\circ$. The cosine is small and positive, so the angle is just under a right angle — the triangle is very nearly right-angled.

$C \approx 87.27^\circ$

12. Drone: $a = 75 \text{ m}$, $b = 110 \text{ m}$, angle between them $C = 132^\circ$. (a) Find c . (b) Why not the law of sines first?

(a)

$$c^2 = 75^2 + 110^2 - 2(75)(110) \cos 132^\circ = 17725 - 16500(-0.6691 \dots) = 28765.6 \dots$$

Because C is obtuse its cosine is negative, so the last term *adds* — and c comes out longer than either leg, which it must.

$$c = 169.6044 \dots$$

$c \approx 169.60 \text{ m}$

Common wrong answer: 81.76 m, from using the reference angle 48° instead of 132° . That flips the sign of the middle term and produces a distance shorter than the longer leg, which no triangle with a 132° angle can have.

(b) *Instructor-judged.* A correct response says the law of sines needs a side together with the angle opposite it, and here the one known angle sits *between* the two known sides, so no such pair exists to start from. The law of cosines is the tool that accepts exactly this arrangement; once c is known, every side has a partner and the law of sines can finish the triangle. Naming the law of cosines without the missing-pair reason is partial credit.